# UM HACKATHON 2026

# Product Requirement Documentation

## Team Free Riders

## AI-Powered Food Waste Redistribution Platform

| TEAM MEMBERS |
| --- |
| 1. JAYDEN FOO LOK HIN |
| 2. VINCENT TEO KAI QI |
| 3. ELVAN SEA MENG JIN |
| 4. KWAN CHUN HOE |
| 5. LIOW YEE SHUEN |

## Table of Contents


1. Project Overview ..... 4
   1.1 Problem Statement ..... 4
   1.2 Target Domain ..... 4
   1.3 Proposed Solution Summary ..... 4
2. Background & Business Objectives ..... 5
   2.1 Background of the Problem ..... 5
   2.2 Importance of Solving This Issue ..... 5
   2.3 Strategic Fit / Impact ..... 5
3. Product Purpose ..... 6
   3.1 Main Goal of the System ..... 6
   3.2 Intended Users (Target Audience) ..... 6
4. System Functionalities ..... 7
   4.1 Description ..... 7
   4.2 Key Functionalities ..... 7
   4.3 AI Model & Prompt Design ..... 7
      4.3.1 Model Selection ..... 7
      4.3.2 Prompting Strategy ..... 8
      4.3.3 Context & Input Handling ..... 8
      4.3.4 Fallback & Failure Behavior ..... 9
5. User Stories & Use Cases ..... 10
   5.1 User Stories ..... 10
   5.2 Use Cases (Main Interactions) ..... 10
      UC-1: Restaurant Donation Listing ..... 10
      UC-2: NGO Claim & Pickup ..... 10
      UC-3: Clarification Workflow ..... 11
   5.3 Use Case Diagram ..... 11
6. Features Included ..... 12
7. Features Not Included ..... 13
8. Assumptions & Constraints ..... 14
   8.1 LLM Cost Constraint ..... 14

8.3.1 Estimated Token Cost per Session: 14
8.3.2 Cost Management Decision: 14
8.3.3 Technical Constraints 14
8.3.4 Performance Constraints 14
8.3.5 User Input Assumptions 14
9. Risks & Questions Throughout Development 16
9.1 Risk Matrix 16
9.2 Design Decision Questions 16

# 1. Project Overview

## 1.1 Problem Statement

Across Malaysia, restaurants and food outlets normally discard large amounts of edible surplus food daily, while NGOs (Non-Governmental Organization) and community kitchens often face irregular food supplies. Currently, the food redistribution efforts rely on fragmented, manual processes such as via WhatsApp messages and phone calls. These result in delayed coordination, mismatched donation, and food expiring before collection. This inefficient persists not because of lack of goodwill but because of poor coordination and ineffective decision making.

## 1.2 Target Domain

AI Systems and Agentic Workflow Automation for Social Impact that specifically addresses food waste reduction (SDG 12) and zero hunger initiatives (SDG 2) through intelligent workflow orchestration.

## 1.3 Proposed Solution Summary

FoodLoop is an AI-powered food redistribution platform that connects restaurants with surplus food to nearby NGOs in real-time. The system leverages Z.AI's GLM (General Language Model) as a central reasoning engine to transform unstructured donation messages into structured, actionable workflows. GLM performs multi-step reasoning to rank NGOs based on food type compatibility, capacity, urgency, distance, and operating hours – replacing manual coordination with intelligent, automated matching. The platform provides real-time listing, AI-driven prioritization, map-based pickup coordination, and transparent tracking of food redistribution activities.

# 2. Background & Business Objectives

## 2.1 Background of the Problem

Malaysia generates approximately 17,000 tonnes of food waste daily, with 24% being still edible. Simultaneously, 1 in 5 Malaysian households faces food insecurity. NGOs like Kechara Soup Kitchen and The Lost Food Project operate with limited staff and manual coordination systems. Restaurant staff, already busy with operations, lack time to find donation recipients. The current workflow is:

*Restaurant has surplus → Calls/texts NGO → NGO busy/closed → Food discarded*

This linear, manual process fails because:

- No real-time visibility of available food
- No intelligent matching based on suitability
- No urgency-based prioritization
- No feedback loop to measure impact

## 2.2 Importance of Solving This Issue

1. **Environmental Impact**: Reducing methane emissions from decomposing food waste in landfills.
2. **Social Impact**: Redirecting 30% of currently wasted edible food to feed vulnerable communities.
3. **Economic Impact**: Restaurants save disposal costs; NGOs reduce food procurement expenses.
4. **Efficiency Impact**: Accelerating donation coordination from hours to seconds, enabling staff to focus on service delivery rather than logistics coordination.

## 2.3 Strategic Fit / Impact

FoodLoop aligns with Malaysia's National Solid Waste Management Policy and the UN's Sustainable Development Goals **(SDG 2: Zero Hunger & SDG 12: Responsible Consumption)**. The platform's AI-native architecture demonstrates how GLM-powered agentic workflows can solve real-world coordination problems beyond simple chatbots – transforming fragmented manual processes into intelligent, scalable systems.

# 3. Product Purpose

## 3.1 Main Goal of the System

To provide an AI-powered coordination engine that transforms unstructured food donation inputs into intelligent, automated redistribution workflows – ensuring surplus food reaches the most suitable NGO before expiry.

## 3.2 Intended Users (Target Audience)

| User Type | Description |
|---|---|
| **Restaurant Staff** | Kitchen managers, owners, F&B outlets |
| **NGO Coordinators** | Food bank operators, community kitchen managers |
| **System Administrators** | Platform operators monitoring impact |

# 4. System Functionalities

## 4.1 Description

FoodLoop operates as an intelligent workflow engine where GLM serves as the central reasoning component. The system accepts unstructured inputs (e.g., "30 packs nasi goreng, 20 packs mi goreng, Restaurant Aliff Sri Petaling"), extracts structured data, performs multi-factor NGO ranking, and orchestrates the entire donation lifecycle from listing to collection confirmation.

## 4.2 Key Functionalities

| Functionality | Description |
|---|---|
| Smart Input Parsing | GLM extracts restaurant name, food items, quantity, pickup time, contact from unstructured text |
| AI-Powered NGO Ranking | Multi-criteria ranking considering food type, capacity, distance, urgency, operating hours |
| Real-time Coordination | Notify top-matching NGOs instantly with priority-based listing display |
| Pickup Route Optimization | Visualize restaurant and NGO locations on Google Maps with optimal routing |
| Stateful Workflow Tracking | Track donation from LISTED → MATCHED → NOTIFICATION IS SENT → ACCEPTED → PICKUP_SCHEDULED → COMPLETED |
| Clarification Loop | Handle ambiguous inputs through intelligent follow-up questions |
| Transparency Logging | Record AI reasoning for each decision (confidence scores, justification) |

## 4.3 AI Model & Prompt Design

### 4.3.1 Model Selection

**Model**: Z.AI GLM

**Justification**:

- Multi-step reasoning capability – Essential for ranking NGOs across multiple constraints (food type, capacity, distance, urgency)
- Structured JSON output – Enables reliable integration with backend systems
- Cost-effective inference – Sustainable for production deployment at scale

### 4.3.2 Prompting Strategy

The approach which the team has chosen to use is a zero-shot prompting strategy with structured instruction approach. Firstly, the model is instructed to extract structured information from the user prompt, such as food type, quantity, expiry & pickup time and others using a crafted system prompt. Then, the information is converted into a standardized JSON format for downstream processing.

We have chosen this strategy as restaurant inputs are short and simple, while also being domain-specific. Instead of using a multi-step prompting approach, this system combines LLM extraction with deterministic post-processing tools to enrich the consistency of data. Additionally, the prompt enforces a controlled response format, allowing the model to request for clarification in cases of incomplete data.

### 4.3.3 Context & Input Handling

The system we designed accepts unstructured natural language input from restaurants and processes it directly through the LLM without prior formatting. Inputs are passed as a single user prompt, relying on the model's instruction-following capability to extract structured data.

While there is no explicit chunking mechanism implemented currently, the system sets a 60 second timeout request, where short to medium donation descriptions remain well within limits. If inputs are abnormally long, the API call raises an error due to timeout, rejecting the input.

To handle ambiguity or incomplete data, the system relies on the LLM to return a structured "incomplete" status, highlighting fields which were incomplete or broken. Then, a follow-up

question for clarification is generated for the user. This ensures that critical information are not silently dropped or ignored to create a complete JSON data.

### 4.3.4 Fallback & Failure Behavior

The system implements multiple safeguards to handle invalid, off-topic or hallucinated outputs:

- Strict JSON Format: The LLM is explicitly instructed to always return a valid structured JSON. After receiving the response, the system removes formatting artifacts and attempts to parse it.
- Automatic Validation Check: If JSON parsing fails due to incomplete or lack of data, the system does not proceed. Instead, it returns an “incomplete” status while sending a clarification request to the user.
- Post-Processing Tool Validation: After receiving a complete response, additional validation and processing tools are implemented using external tools (time parser, quantity extraction, food classifier) to reduce the impact of minor inconsistencies or hallucinated formats.

# 5. User Stories & Use Cases

## 5.1 User Stories

| User | Story |
|---|---|
| **Restaurant Staff** | "As a restaurant manager, I want to type a quick message about surplus food so that I can donate it without complex forms." |
| **Restaurant Staff** | "As a busy kitchen worker, I want the system to understand my informal language so that I don't need to follow strict formats." |
| **NGO Coordinator** | "As an NGO coordinator, I want to see only relevant food listings so that I don't waste time on unsuitable donations." |
| **NGO Coordinator** | "As a food bank operator, I want to know why an NGO was prioritized so that I understand the matching logic." |
| **Volunteer Driver** | "As a pickup volunteer, I want optimized routes so that I can collect multiple donations efficiently." |
| **System Admin** | "As a platform operator, I want to see AI decision logs so that I can audit matching fairness." |

## 5.2 Use Cases (Main Interactions)

### UC-1: Restaurant Donation Listing

1. Restaurant staff types donation message (e.g., "20 packs nasi lemak, eat before 9 pm, Kedai Mamak SS15")
2. GLM extracts structured data
3. System validates completeness → asks clarification if needed
4. AI ranks suitable NGOs
5. Top NGOs receive real-time notifications

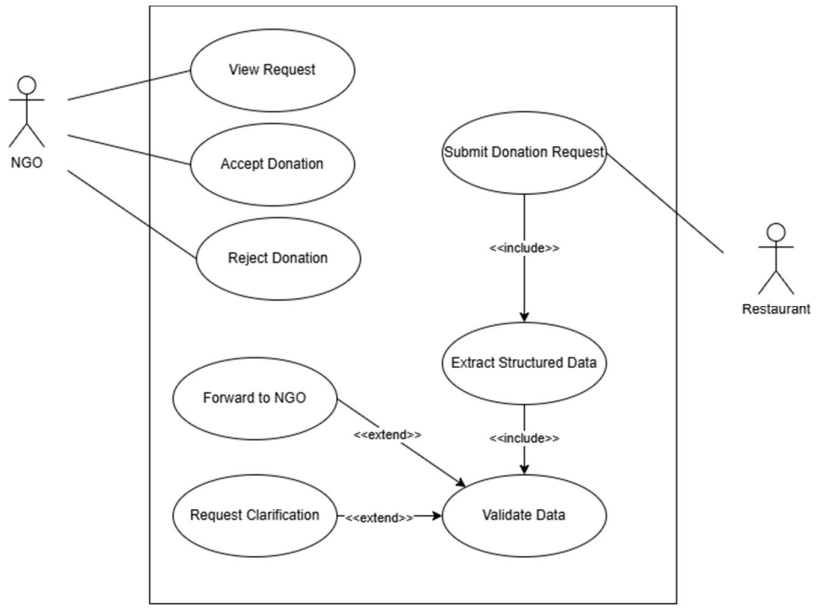
### UC-2: NGO Claim & Pickup

1. NGO views prioritized listings on dashboard
2. NGO claims available donation
3. System generates pickup instructions with the map route
4. Volunteer collects food
5. NGO confirms completion → closes loop

### UC-3: Clarification Workflow

1. User provides incomplete input ("I have food")
2. GLM detects missing fields
3. System asks specific clarification question
4. User provides missing information
5. Original context preserved and workflow resumes

## 5.3 Use Case Diagram




There are two actors in this use case, which is the NGO and Restaurant. The Restaurant may begin by submitting a donation request, in which structured data will be extracted and validated from the donation request by the System. Then, based on whether the data is complete or incomplete, two use cases may occur. The first use case is where the data is complete and it will be forwarded to the NGO. Secondly, assuming that the data is incomplete or missing certain fields, a clarification request will be sent.

Aside from that, the NGO actor has three use cases. They are able to view the completed donation request which has been sent by the system. Then, they may choose to accept or reject the donation.

# 6. Features Included

| Feature | Description | AI Dependency |
|---|---|---|
| **Unstructured Input Parsing** | GLM extracts structured data from free text messages | Core AI |
| **Intelligent NGO Ranking** | Multi-criteria ranking based on food type, capacity, distance, urgency | Core AI |
| **Stateful Workflow Engine** | Track donation lifecycle (LISTED → MATCHED → NOTIFICATION IS SENT → ACCEPTED → PICKUP_SCHEDULED → COMPLETED) | System logic |
| **Clarification Questions** | AI-generated follow-up for missing/ambiguous fields | Core AI |
| **Real-time Notifications** | Push notifications to top-ranked NGOs | System logic |
| **Map-based Routing** | Google Maps integration for pickup visualization | External API |
| **Transparency Logging** | Record AI reasoning and confidence scores | Core AI |
| **Conversation Context** | Maintain history for multi-turn clarification | System logic |
| **Food Safety Validation** | Reject donations with expiry before pickup | Core AI |
| **Quantity Extraction** | Extract numeric quantities from text using regex tool | Helper tool |
| **Time Normalization** | Convert human time expressions to ISO format | Helper tool |
| **Food Classification** | Categorize food (cooked_meal, bakery, raw, etc.) | Helper tool |

# 7. Features Not Included

| Feature | Reason for Exclusion |
|---|---|
| **Real-time GPS tracking** | Complexity exceeds hackathon scope |
| **Payment processing** | Not required for donation platform |
| **Multi-language UI** | MVP focuses on English/Malay bilingual support |
| **Volunteer scheduling system** | Future enhancement post-MVP |
| **Analytics dashboard** | Basic logging only for hackathon |
| **Mobile native apps** | Web-based responsive PWA only |
| **Blockchain verification** | Over-engineering for problem scope |
| **NGO rating system** | Requires extensive user base |

# 8. Assumptions & Constraints

## 8.1 LLM Cost Constraint

### 8.3.1 Estimated Token Cost per Session:

- Average user session: 500-800 tokens (input + output)
- NGO ranking call: 300-400 tokens
- Total per donation: ~1,200 tokens

### 8.3.2 Cost Management Decision:

- **Caching strategy:** Frequent NGO profiles cached, only changed fields re-submitted
- **Prompt optimization:** System prompt includes only essential NGO fields (location, capacity, preferences) – not full NGO descriptions
- **Result limiting:** Return top 5 ranked NGOs only, not full list

### 8.3.3 Technical Constraints

| Constraint | Limitation |
|---|---|
| **API Rate Limits** | Z.AI GLM: 60 requests/minute maximum |
| **Input Size** | 2000 character limit per prompt |
| **Response Time** | LLM inference: 2-5 seconds per request |
| **Concurrent Users** | MVP supports 50 concurrent sessions |
| **Data Persistence** | Firebase Firestore free tier: 1GB storage |

### 8.3.4 Performance Constraints

- **Generation limit:** Maximum 3 ranking requests per donation (prevents abuse)
- **Notification delay:** Real-time not guaranteed; 5-10 second typical latency
- **Map loading:** Google Maps API quota: 28,000 requests/month free tier

### 8.3.5 User Input Assumptions

- Users have basic smartphone/internet access
- Restaurant staff can type/voice basic English or Malay

- NGOs have staff to monitor dashboard during operating hours
- Pickup volunteers have personal vehicles for collection

# 9. Risks & Questions Throughout Development

## 9.1 Risk Matrix

| Risk | Probability | Impact | Mitigation Strategy |
|---|---|---|---|
| **GLM hallucinates NGO ranking** | Medium | High | Schema validation + confidence score threshold; reject low-confidence rankings |
| **Expired food accepted** | Low | Critical | Dual validation: GLM + time_parser tool; hard rejection if expiry < pickup |
| **NGO capacity exceeded** | Medium | Medium | Track active claims per NGO; prevent over-assignment via Firestore counters |
| **API downtime** | Low | High | Graceful error with retry; NO fallback (proves AI necessity) |
| **Location geocoding fails** | Medium | Low | Fallback to distance calculation via Google Maps API only |
| **Malay language confusion** | Medium | Medium | System prompt includes bilingual examples; field extraction handles common Malay terms (nasi, roti, sayur) |

## 9.2 Design Decision Questions

1. Why is there no fallback if GLM fails?
    - Answer: Problem statement requires that "If GLM component is removed, system loses ability to coordinate." Rule-based fallback would violate this requirement.
2. Why rank only top 5 NGOs?
    - Answer: Token cost optimization; restaurants don't need to see all 50 NGOs; reduces cognitive load.